ImageVault: A Decentralized Blockchain-based Image Management Platform

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Abstract. In today's digital age, most existing image storage systems are built on centralized cloud infrastructure or partially decentralized systems. These systems expose sensitive personal and corporate images to risks such as privacy (unauthorized access, single point failure) and integrity (data tempering). While blockchain technology primarily mitigates these risks, it focuses on hash-based security and immutability. To inspire with this technology, this work proposes a decentralized and distributed image storage platform to secure images and preserve their privacy and integrity. It uses the InterPlanetary File System and blockchain technology. To protect ownership and access privileges of images, the platform integrates IPFS's content-addressed storage and blockchain's immutability. This Ethereum-based solution ensures seamless owner approvals, access requests, and image uploads without intermediaries. To offer strong security and trust, standout features include transparent access restriction, local encryption, and immutable transaction recording. This research shows how blockchain technology can be integrated with IPFS to enhance user trust in managing sensitive digital content, protect private photographs, and eliminate the need for centralized storage. The proposed system addresses the limitations of centralized and semi-decentralized image storage systems. This proposal is capable to secure all types of images, which means it is not limited to images from a specific domain.

Keywords: Blockchain · Ethereum · InterPlanetary File System · MetaMask · Smart Contracts.

1 Introduction

In today's digital era, most individuals and organizations utilize cloud platforms for their daily tasks. Google Drive [12], Dropbox [10], and AWS [1] are the most

popular cloud platforms, and have significant market share. These platforms are commonly used to store and share images among peers. These digital images may be personal photos or sensitive corporate images. These platforms offer convenience and flexibility to users, but their centralized architecture introduces significant security concerns over the stored images. These concerns are vulnerability, susceptibility to data breaches, unauthorized access, service disruptions, and overdependence on third-party providers [9]. In the past, many reported incidents have been witnessed, where users have lost control over digital content/images stored on cloud platforms. These incidents typically lack accountability and transparent auditing mechanisms in these platforms [29]. Even with advanced encryption and access control mechanisms, centralized architecture remains a critical weakness [23] [31]. These centralized systems are fully controlled and managed by a central authority. This central authority manages keys, enforces policies, and maintains infrastructure, which creates a single point of failure (SPOF) and trust bottlenecks. In many cases, server logs are also altered, and the platforms discard access requests. Such incidents reduce transparency and trust between the user and the platform [5] [30].

Over the past decade, Decentralized Identity (DID) systems have gained attention as an alternative to centralized systems. These systems introduce the concept of self-sovereign identity (SSI), which enables users to manage their own identity verification. SSI eliminates the role of central authorities from the verification process [6]. It also reduces reliance on third-party authentication and enhances privacy. However, these DID systems are unable to address tamper-resistant access control, encrypted data distribution, and decentralized storage issues. These systems primarily manage identity but do not provide immutable logs, extensible storage, and verifiable access control mechanisms [24].

With recent developments, Blockchain Technology (BCT) has shown its capability to address these issues. It offers an effective solution for record-keeping applications. It provides a decentralized, immutable, transparent and permanent record-keeping ledger [19] [7] [2]. In blockchains (BCs), smart contracts (SCs) are responsible for automating and enforcing compliance in the system. Still, the high costs and scalability limitations of storing large files directly on-chain make this technology impractical. To overcome this limitation, BC-based Platforms (BCP) can use the onm-chain and off-chain concepts, which reduce costs and improve scalability. In this, BC stores only cryptographic hashes and metadata on-chain, while the actual files remain off-chain. The InterPlanetary File System (IPFS) [28] is a peer-to-peer content-addressed file system. It provides off-chain storage in BC, which can be used as a complementary solution. It provides decentralized storage for large files. This stored file is accessible/identifiable through a unique content hash (i.e, CID) rather than location-based URLs. It improves reliability and availability in the platform [4].

The integration of BCP, BCT, and IPFS improves security, traceability, and immutability on the platform/system. However, many challenges still persist in these platforms/systems. These are complex access control, encryption overhead, and performance trade-offs [27]. To address these challenges, this work proposes

a decentralized, BC-based image management platform, which is named ImageVault. The proposed platform uses Ganache [11] and Truffle Suite [32] for the development and deployment of the system. Both tools provide the development platform for developers to develop, deploy and test SC in the Ethereum BC test scenario. Ganache provides a private Ethereum BC environment for testing, while Truffle manages automated testing, migrations, and compilation. Together, these tools ensure a robust and reliable evaluation of the developed platform. This proposal offers strong generalization capabilities, which make it a powerful platform capable of securing any type of image. Unlike existing systems that are limited to specific domains such as healthcare or education. The proposed ImageVault does not restrict to any single application area.

The proposed system addresses the security and privacy limitations of centralized systems, including a single point of failure, lack of transparent access governance, and reliance on trusted third parties to enforce ownership and access policies. It also improves the visibility of image owners (i.e., how their images are accessed) and their control over access revocation. It provides an immutable audit trail of image usage to owners. In short, ImageVault provides a decentralized, owner-controlled image storage platform that combines encrypted off-chain storage with blockchain-enforced access governance, delivering privacy preservation, tamper resistance, and verifiable accountability.

The rest of the chapter is organized as follows. Section 2 discusses the related literature works. Section 3 presents the proposed system architecture. Section 4 discusses system configuration and workflow. Section 5 determines the implementation of smart contracts, describing the results obtained and computing the performance of the proposed system. Lastly, concluding remarks are detailed in Section 6.

2 Related Work

Early developments in the domain of image management platforms primarily focus on encrypting files and storing them on a cloud platform, with access controlled by standard authentication methods. However, the rise of blockchain inspired a change in the platform. The BCT and distributed file systems (DFS) now offer the foundation for a secure, tamper-resistant, and decentralized storage platform for digital assets, especially sensitive images. These platforms ensure secure and transparent access policies, as well as immutable logs of stored information and images. All these features and policies are automated via the SCs.

IPFS provides an effective off-chain storage solution, which enables image files to be stored outside the BC. This prevents the blockchain network (BCNw) from becoming overloaded and helps maintain performance. Only the content hashes and the access-control logic are kept on-chain. It is benefiting from security and integrity in BC. This architecture serves as the foundation for reviewing existing research that explores off-chain storage and encryption techniques for digital images using DC.

In paper [28], the authors propose a secure storage model that combines BCT with IPFS for distributed file storage in a decentralized environment. In this, image files are stored in IPFS, while BCT manages access and ensures protection. The proposed model aims to ensure the security of medical images without depending on third-party centralized cloud system and eliminates the limitations associated with centralized storage. Similarly, in [3], the authors propose a BC-based framework to store medical images. This framework also uses IPFS to provide decentralized storage, and its corresponding unique hash is stored in the BCNw. This work also emphasizes on the limitations of BC, such as performance bottlenecks when transaction throughput increases. The proposed framework also needs additional layers of security and must be tested in real-world healthcare environments.

In [17], the authors propose a secure storage system for chest X-ray images, which is built on BC with a Kaggle-based implementation. In a system, an SC regulates role-based access permissions and authenticates medical images. Their initial findings indicate that the BC-based approach adds around 5% performance overhead. Another study [33] presents a hybrid framework that combines BC with fractional DCT (Discrete Cosine Transform) to store and retrieve medical images. The proposed framework improves data sharing, enhances security, supports efficient retrieval, and offers energy-efficient computation, contributing to overall sustainability.

Work [15] integrates Ethereum BC with IPFS to develop a distributed patient-centric image management (PCIM) system. This system provides a secure storage and sharing platform for medical photos. The proposed decentralized and distributed system enhances data privacy, minimizes maintenance costs, and reduces reliance on central cloud infrastructure. Its maintenance cost is minimized because it does not require centralized cloud infrastructure. PCIM offers a robust and distributed access control system that uses a patient-centric access control protocol. An Ethereum-based SC governs this protocol. In PCIM, IPFS ensures the security, integrity and global accessibility of the stored medical images. Testing of the PCIM on the Ethereum testnet confirms the feasibility and viability of the architecture. Similarly [20] proposes a patient-centric report and image management (PCRIM) system using the same combination of Ethereum BC and IPFS. The proposed system provides a storage and sharing platform for medical data, eliminating the need for centralized infrastructures. This proposal addresses the high maintenance costs, restricted scalability, and privacy concerns associated with cloud-based systems. The proposed system uses an Ethereum SC to implement the decentralized access control policy in the system. This policy is also known as the Patient-Centric Access Control Protocol (PCACP). In the system, IPFS provides decentralized storage for medical images and their metadata. It ensures data integrity and worldwide accessibility of data. The Ethereum test net prototype implementation indicates that the PCRIM architecture is feasible, safe, and effective for handling confidential medical data in a decentralized setting.

In paper [21], the authors propose a BC-based digital copyright management system (DCMS). It addresses the issues associated with centralized copyright registration systems. It integrates perceptual hashing techniques with the IPFS. DCMS addresses challenges such as high service prices, long processing times, and tamper risk. It uses IPFS to store photos and copyright information on a distributed network securely. This BCT-based framework protects the data integrity and transparency in digital assets. It utilizes perceptual image hashing with content similarity to confirm and validate image ownership. It is a peer-to-peer (P2P) network that ensures digital assets are protected, safe, and free from distribution by third parties. In another paper [8], the authors present a BC-integrated architecture for digital copyright protection. This architecture primarily enhances digital copyright protection and source verification for images that are stored on IPFS. This development integrates the blind watermarking, digital signatures (DSigns), and BCT. These fundamental technologies ensure the validity, integrity, and traceability of digital images/media. Here, DSigns ensures data protection and uploader authentication, while BCT provides a decentralized, immutable, transparent, and traceable ledger for images. The blind watermarking embeds the undetectable ownership/copyright information into images. The architecture of the proposed system is a dual-layer Distributed Hash Table (DHT) indexing structure. It includes both global and local index layers. This dual-layer structure improves retrieval efficiency while minimizing search latency and network overhead. This work also demonstrates and validates the system's scalability and efficiency in addressing copyright and ownership concerns related to IPFS.

In paper [16], the authors raise concerns about storage due to the rapid growth of transaction data in BC systems. They present a BC-based storage mechanism using IPFS to tackle the concerns. The proposed mechanism stores transactions on IPFS and records the generated IPFS hash in the block of the BC. This approach preserves data availability and immutability while significantly lowering the block size. This proposed architecture promises both secure and efficient access to transaction data by using a content-addressed storage method rather than traditional location-based addressing. The framework is implemented with Anaconda Python, Flask, and IPFS. The viability and performance of the framework in safely handling picture transactions are evaluated with the implementation. In another work [18], authors discuss a Distributed Image Storage Protocol (DISP) to overcome the storage issues in BCT. The current storage reduction techniques frequently improve calculation time, but they compromise the fault tolerance. DISP merges distributed databases with IPFS to enhance storage efficiency and minimize computational costs. An experimental simulation verifies its reliability. This proposal breaks the visual proof into smaller and distinguishable pieces. These pieces are stored among several nodes using the Distributed Pooling (DP) algorithm. These pieces can be accurately rebuilt using the Inverse Distributed Pooling (IDP) algorithm during the authorization. This distributed strategy may benefit large-scale commercial and industrial projects that require effective and secure storage management. It op-

timizes resource utilization, maintains data integrity, and significantly improves scalability.

All these proposals strengthen the blockchain ecosystem for storing different types of image records. All these are summerized in Table 1.

3 Proposed System Architecture

The proposed system primarily addresses the challenges of the traditional centralized storage systems. It integrates IPFS with BCT to develop a decentralized framework. This proposed system aims to develop a privacy-preserving system. This system is capable to store the encrypted images, provide an image-sharing platform, and securely manages access control. The system ensures that ownership and control of images remain with users rather than a third-party provider, unlike centralized cloud systems. In this framework, a user first encrypts the images at the client side, then uploads the encrypted images to IPFS. This setup ensures confidentiality in the system. In this architecture, only essential metadata information, such as the CID and access policies, is stored on the on-chain BC. An image on-chain verification request is logged through the user interface, and its approval initiates the secure key exchange between the user and the requester. This key exchange is made possible via public key cryptography through a Node.js-based encryption API.

Overall, this proposed framework enhances user sovereignty, strengthens access control, and enables transparent auditing in the system. This system integrates several decentralized technologies, including IPFS, SCs, DIDs, Zero-Knowledge Proofs (ZKPs), and BCT. IPFS provides a distributed and resilient file storage; SC enforces and automates access policies; DID and ZKP provide a privacy-preserving authentication mechanism; and BCT provides an immutable ledger for logging access events.

This system is explicitly designed to eliminate SPOF in the system. It preserves user control data and encryption keys, ensuring that every access request and data modification attempt is permanently recorded in the ledger. Here, the user encrypts the image files using the AES-256 algorithm and then uploads the encrypted file to IPFS. In this, encryption keys are also secured using the public keys of authorized users through encryption. This enhances system security. When a requester submits an image access request through the user interface, the decentralized identity system first verifies the requester's identity. After the request is validated and approved, the requester retrieves the file from IPFS. The requester then decrypts the retrieved image locally using the associated CID and the encrypted key. Every image access attempt, both successful and denied, is recorded on the BC with a hashed user identity, timestamp, and CID reference. Additional safeguard measures of the system include stake- or proof-based verification against Sybil attacks. It secures key wrapping and reduces the risk of key leakage, as well as pre-deployment auditing of SCs to prevent exploits.

This proposed framework defines three types of user based on their roles and responsibilities. These are Image_Owner, Image_Requester, and Image_Administrator.

Table 1: Summary of studied works.

Ref.	Main Objective / Contribution	Technologies Used	Key Features / Methods	Limitations / Remarks
[28]	Secure decentralized storage of medical images	BCT, IPFS	Files stored in IPFS; BCT ensures access control and protection	Removes third-party dependency; not tested in real-world healthcare
[3]	Blockchain framework for storing medical images	BCT, IPFS	Image hashes stored on blockchain; decentralized storage	Scalability issues due to high transactions; needs stronger security layers
[17]	Secure chest X-ray storage using blockchain	BCT, SC	Role-based access control; authentication via SC	Low overhead ($\approx 5\%$); initial findings only
[33]	Efficient medical image storage and retrieval	BCT, Fractional DCT	Data sharing, compression, and retrieval efficiency; sustainable energy use	Requires broader validation
[16]	Blockchain storage optimization via IPFS	BCT, IPFS, Flask, Python	Stores transactions on IPFS; hashes in blockchain to reduce block size	Effective for transaction data; not specialized for medical domain
[15]	Patient-Centric Image Management (PCIM)	Ethereum, IPFS, SC	Decentralized image sharing; patient-centric access control	Prototype tested only on Ethereum testnet
[21]	Blockchain-based digital copyright management	BCT, IPFS, Perceptual Hashing	Decentralized copyright verification; image similarity-based ownership	Limited evaluation of scalability and network overhead
[20]	Patient-Centric Report and Image Management (PCRIM)	Ethereum, IPFS, SC	Distributed medical data storage; decentralized access policy	Prototype validation only; similar to PCIM with report integration
[8]	Digital copyright protection for images	BCT, IPFS, Blind Watermarking, DSigns	Combines watermarking and DSigns; DHT indexing for efficiency	Focused on copyright; high complexity due to dual-layer DHT
[18]	Distributed Image Storage Protocol (DISP)	BCT, IPFS, DP & IDP Algorithms	Splits images into fragments; distributed storage and reconstruction	Computational complexity; simulation-based validation

An Image_Owner uploads the encrypted content and manages access rights. An Image_Requester submits access requests to view specific images, and an Image_Administrator oversees or audits the system. At the backend, the system uses the ECCrypto library to implement the Elliptic Curve Integrated Encryption Scheme (ECIES). It ensures secure key generation, encryption, and signature verification. The Image_Owner uses a key pair to decrypt the image. This pair contains the decryption key and the public key of the requester. This key combination ensures that only the correct recipient user can access the image. The correct recipient always has a matching private key. This process guarantees end-to-end confidentiality and non-repudiation in the framework.

In IPFS, each file is uniquely identified by its hash value (i.e. CID). This CID ensures integrity in verification, resilience against censorship, and SPOF [15]. It also enables content addressing, redundancy, and durability in the framework. This framework provides a user-friendly front-end interface to users. It is built on Angular and facilitates users to encrypt and upload images, manage access requests, and decrypt the images on request [26]. This front-end interacts with SCs via Web3.js and ethers.js, and integrates with cryptographic APIs for key management.

This on-chain and off-chain storage mechanism provides a scalable, cost-efficient framework for users. In this framework, BCNw (i.e., on-chain) stores only metadata and access logs of images, while the images themselves are stored in IPFS (i.e., off-chain) in encrypted format [22]. It reduces gas costs and supports large-scale deployment. Furthermore, the modular framework can extend beyond the image storage to other sensitive digital assets. These digital assets can be educational certificates, legal documents, property records, supply chain records, and medical records. This hybrid on-chain/off-chain concept is designed to balance security, transparency, and scalability. Computationally expansive and large data operations are handled off-chain, while critical access-control and audit functions are enforced on-chain. The architecture of the proposed system is illustrated in Figure 1.

The proposed ImageVault system is clearly distinguished from prior blockchain –IPFS approaches through its integrated privacy-preserving architecture that unifies encryption, access delegation, key management, and audit logging via smart contracts. Unlike earlier solutions with static CID-based permissions, ImageVault offers dynamic, fine-grained access control with support for revocation and temporary access tokens. It further introduces an end-to-end key-rotation mechanism that eliminates the need to re-upload content to IPFS, along with a decentralized integrity and privacy auditing process that goes beyond basic on-chain hashing.

4 System Configuration and Workflow

The configuration and operational workflow of the proposed system is presented in this section. This discussion includes the BC environment setup process, the

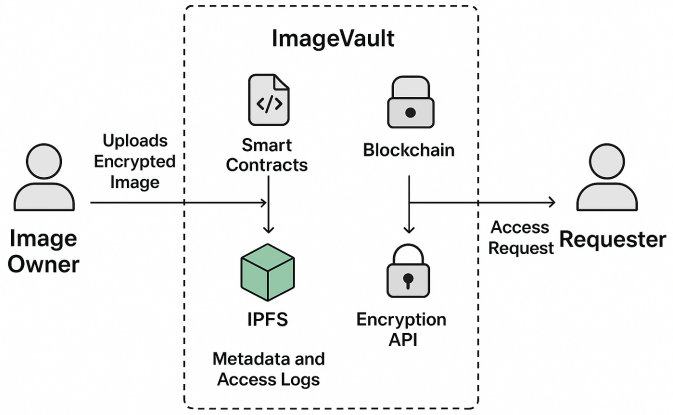

Fig. 1: Architecture of proposed ImageVault.

IPFS integration process, and a mechanism for securely storing images, along with their access control mechanisms.

4.1 Environment Setup

The proposed system integrates BCT and IPFS. It provides a decentralized image storage framework to users. This framework features an Angular-based user interface that enables users to interact with the system. In the framework, an SC is responsible for ownership, access control, and metadata management. This SC is developed in Solidity language, deployed in Ethereum through the Truffle Suite and locally tested using Ganache. In the framework, MetaMask is responsible for wallet management and transaction security. It also ensures user authentication and secure interaction between users. In the framework, the user first uploads the encrypted images to IPFS, and then their CID are uploaded to the BC network as an immutable record. The frontend of the framework ensures secure communication between the BC and users via Web3.js or Ethers.js. It also ensures responsive and secure contract event handling in the framework. Collectively, all these components develop a cohesive and secure framework that supports BC-based access control, encryption, and decentralized image storage.

4.2 Integrated Image Storage and Access Workflow

In ImageVault, an image owner can interact with the framework through the Angular-based frontend application. During image upload, the image owner first encrypts the image locally and then uploads this encrypted image to the IPFS network. IPFS provides a unique CID for that stored image. This CID, along with

the metadata and access permissions, will be stored in the BC as a transaction. This complete process is controlled and automated via an SC. All transactions are authenticated through MetaMask to preserve the integrity of the ownership. An image requester can interact with the frontend and send an image access request to the framework. This request creates a corresponding entry in the BC. This image access request is forwarded to the image owner. The image owner is responsible for approving or denying the request. After successful approval, an authorized user retrieves the image from IPFS using its CID and decrypts it using their decryption keys.

The SCs are developed in the Truffle suite and tested using the Ganache frameworks. It handles image registration, access control, and event logging. The integration of SCs with Web3.js or Ethers.js enables a secure, real-time interaction between the frontend and BCNw. The developed system is deployed from the local Ganache network to Ethereum test networks. It incorporates IPFS pinning and encrypted key exchange to ensure reliability, decentralization, and long-term data availability. This complete workflow of the ImageVault system is illustrated in Figure 2.

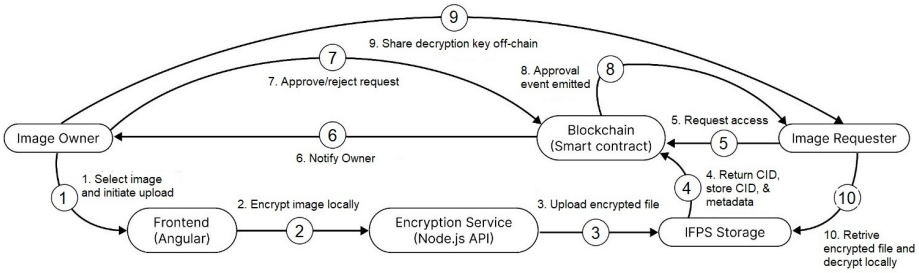

Fig. 2: Diagram of the image storage and access workflow in ImageVault.

4.3 Blockchain Functionality in the Image Storage System

This framework combines BC and IPFS to form a decentralized and tamper-resistant framework for image storage, authorization, and auditing. The BC maintains immutable records of image metadata, transaction logs, ownership details, and access permissions. It ensures data transparency and integrity in the system [13]. The framework is simulated in a private Ethereum environment using Ganache. It allows developers to test and debug SCs, monitor wallet transactions, and analyze gas consumption without relying on public networks or incurring transaction costs [13].

4.4 Consensus Mechanism in ImageVault

The proposed framework is developed over a private and permissioned BC. It uses the Proof-of-Authority (PoA) consensus mechanism (CM). In this, pre-approved validators create blocks and validate transactions [13]. These pre-approved validators can be institutional operators or trusted network nodes. This CM is particularly best suited for private, permissioned environments. In private BC, all participants are known and verified. PoA offers advantages such as high throughput, low latency, and minimal computational overhead. These features make it ideal for business and research users who require efficient performance and controlled participation [14]. However, PoA introduces a degree of centralization in the BCNw, as a limited number of validators control block validation. This centralization reduces fault tolerance and decentralization [25]. Despite this trade-off, PoA provides a practical balance between security, manageability, and operational efficiency [16]. In the context of the proposed system, PoA is an appropriate choice since all validators are trusted and predefined, ensuring network integrity and reliability.

5 Results

The aim of the proposed framework is to develop a secure and decentralized image storage with fine-grained access control. It integrates BC with the IPFS for that. Although the system has not yet been deployed in real-world environments, its performance has been evaluated in a test environment. For that, we use the Truffle Suite, Ganache, MetaMask, and Web3.js. The various screenshots of this development are shown in Figure 3. It shows user-friendly GUIs, as well as local deployment of Smart Contracts using Ganache, and its block generation and transaction records.

In the test environment, the average execution time of an SC is less than 3.0 seconds on the local BC. It ensures the timely management of access requests. The retrieval time of an image from the IPFS is less than 2.0 seconds for files smaller than 5.0 MB. It demonstrates low latency for regular image applications. The average time for encryption and decryption operations is less than 1.0 second. These operations are implemented using the ECC algorithm in the framework. It ensures the secure processing of images without noticeable delays for end users.

This proposed system also ensures that no sensitive image data is stored on-chain in plaintext. All images are encrypted locally and then stored over the BC network. This on- and off- chain concept improves the scalability of system and avoids the limitations associated with large files. The BC provides immutability in the framework. It maintains a transparent audit trail for every access request and approval, thereby minimizing the risk of unauthorized modifications. In this platform, images are encrypted prior to upload, and decryption keys are shared securely in off-chain. This process ensures confidentiality in the framework even when operating over the public IPFS network. Table 1 summarizes the per-

formance metrics of the proposed ImageVault application (dApp), including SC execution time, image retrieval latency, and encryption and decryption overhead.

Table 2: System performance of ImageVault.

Parameter	Value	Description
Smart Contract Execution Time	2.7 seconds	Time required in approval/rejection of transaction in Ganache
IPFS Retrieval Time (File $\leq$ 5 MB)	1.9 seconds	Time required to fetch encrypted image from IPFS
Encryption Time (ECC)	2.3 second	Time required to encrypt image before upload
Decryption Time (ECC)	0.5 second	Time required to decrypt image locally
BC Storage Overhead	Minimal (CID only)	Only storing metadata & CID on-chain to avoid large file storage

6 Conclusion and Future Direction

This paper presents the concept, functionality, and initial evaluation of ImageVault, a decentralized platform for secure image storage. This platform utilizes blockchain for controlled sharing, IPFS for distributed file storage, and client-side encryption for data privacy. The system encrypts the images locally before uploading them to IPFS, and stores only the metadata and access policies on the Ethereum blockchain. In the system, the smart contract manages the ownership, access control, and transaction log. The public-key cryptography technique enables secure key exchanges between users. The Angular-based front end, integrated with web3.js or ethers.js, provides an intuitive interface for managing uploads, approvals, and access requests. This proposed system demonstrates the practical integration of decentralized technologies to develop privacy-preserving, transparent, and scalable image management systems which is suitable for various real-world applications. ImageVault offers a decentralized, owner-centric image storage platform that integrates encrypted off-chain storage with blockchain-enforced access control, ensuring privacy preservation, tamper resistance, and verifiable accountability. This proposed system requires a more in-depth evaluation in a real-world scenario and requires comparison with existing systems. A formal analysis of the results is also required to demonstrate the security and robustness of the system. The current work lacks a detailed discussion of the threat model and potential weaknesses of the system.

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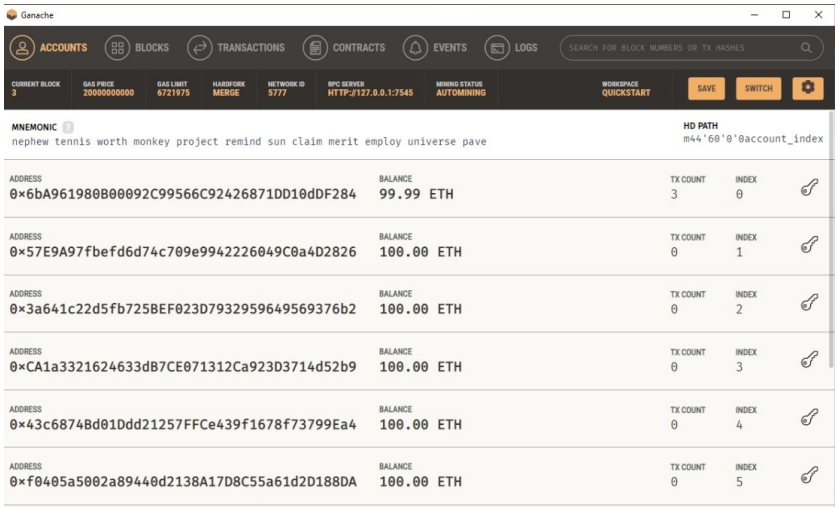
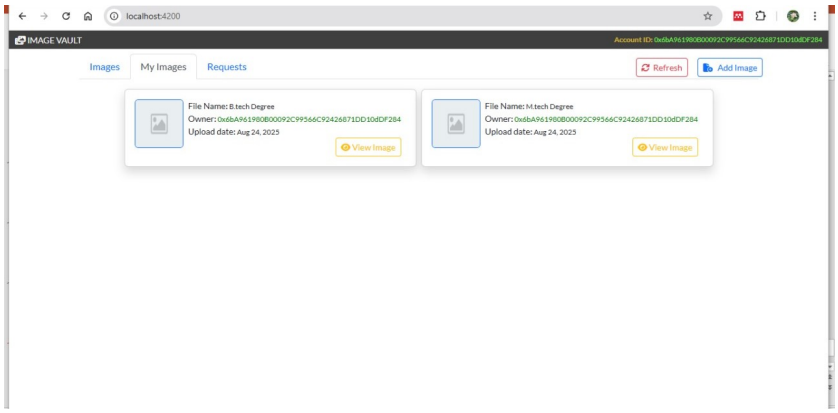
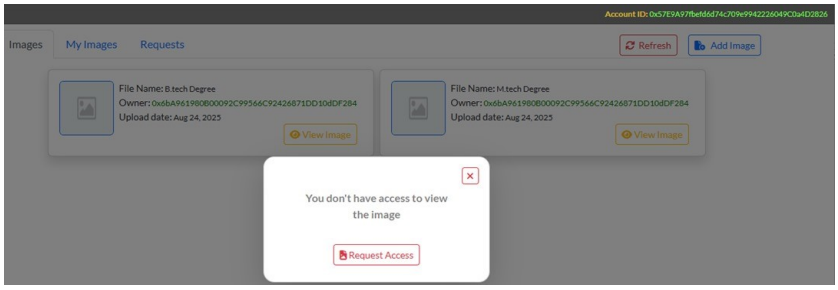


(a) Transaction records of Ganache.



(b) ImageVault repository.



(c) Unauthorized access of ImageVault repository.

Fig. 3: Various screenshots of ImageVault development.

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